

Sustained Deep-Network Binding Is a Correlate of Accessibility Concept Emergence: Evidence from the Pythia and GPT-2 Model Suites

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Abstract

Sustained deep-network binding of accessibility compounds appears to be a necessary structural condition for behavioral capability — present in every model that correctly defines core concepts, absent in every model that fails. We use Web Content Accessibility Guidelines (WCAG) as our test domain because accessibility represents a specialized, low-frequency domain in web-scale training data — a small, well-defined vocabulary with unambiguous answers and direct relevance to real-world tooling decisions, making it an unusually clean lens for studying emergence: concepts are concrete enough to evaluate and rare enough to show scale sensitivity. Using the Pythia suite (160M–6.9B) and GPT-2 (small–XL) with TransformerLens, we investigate not just what models know but how that knowledge is encoded internally. All models show compound binding in early layers; the differentiating factor is whether that binding persists to late network layers. Screen reader, skip link, and alt text emerge behaviorally at ~2.8B; WCAG first appears at 6.9B; Accessible Rich Internet Applications (ARIA) exhibits fluent wrongness at every scale tested — producing confident, plausible expansions that are consistently incorrect. Models prefer correct definitions before they can produce them, and a declarative-evaluative gap persists even at maximum scale: models that correctly define accessibility concepts cannot reliably identify violations in code.

Introduction

Digital accessibility guidelines represent a narrow, specialized domain in web-scale training data. Terms like WCAG, ARIA, and alt text appear far less frequently than general web vocabulary, and their correct usage requires understanding both syntax and intent. This makes accessibility an unusual

test case for studying emergence because the concepts are concrete enough to evaluate and rare enough to show scale sensitivity.

Most emergence research uses broad capability benchmarks like arithmetic, chain-of-thought reasoning, and multilingual translation. These are useful but can be noisy. Accessibility concepts offer something different: a small, well-defined vocabulary with unambiguous answers and direct relevance to real-world tooling decisions.

Binding Depth and Behavioral Emergence Follow the Same Shape

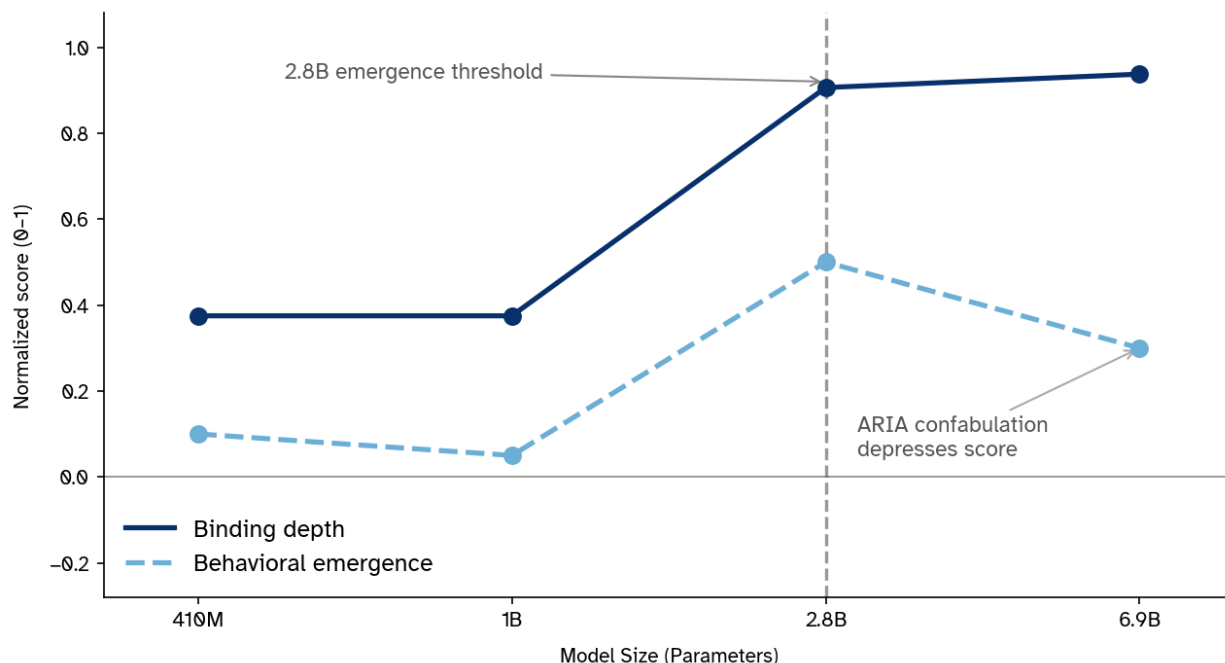


Figure 1: Binding depth and behavioral emergence score rise at the same 2.8B parameter threshold across the Pythia model suite. ARIA scores negative at 1B--6.9B reflecting fluent confabulation rather than absence of response.

The findings are specific: accessibility knowledge emerges at 2.8B parameters for core concepts, later or not at all for acronyms, and evaluative capability lags behind declarative knowledge even at maximum scale tested. Attention pattern analysis shows mechanistic encoding of compound terms before full behavioral competence appears — internal representation that is not reducible to output accuracy. Replication across the GPT-2 model

suite confirms that the core patterns — recognition preceding generation, deep-network binding persistence as a correlate of emergence, and the declarative-evaluative gap — hold across architectures and training corpora.

A preliminary version of these findings appeared as a blog post (Salas, 2026); we present the complete experimental results with extended analysis and cross-architecture replication.

Related Work

Emergence in language models. Wei et al. (2022) established that language models exhibit emergent abilities — capabilities that appear sharply at specific parameter scales rather than improving gradually. We treat accessibility concept acquisition as a domain-specific case study of emergence, using the Pythia suite’s controlled architecture to isolate scale as the variable of interest.

Knowledge representation in transformers. Geva et al. (2021) demonstrated that feed-forward layers in transformers function as key-value memories, storing factual associations that attention heads retrieve and route. Subsequent work by Yao et al. (2024) mapped knowledge circuits in GPT-2 Medium using factual recall tasks, noting explicitly that their findings “may not be universally applicable to all knowledge domains.” We examine whether domain-specific, low-frequency knowledge follows the same patterns.

Attention as an interpretability tool. Clark et al. (2019) showed that attention patterns in BERT encode linguistically meaningful relationships, establishing attention visualization as a valid method for probing internal representations. We apply the same approach to compound accessibility terms, asking whether models bind multi-token concepts as units.

Mechanistic interpretability. Olsson et al. (2022) characterized induction circuits and previous-token heads in transformer models, providing the framework used here to interpret early-layer binding patterns. The distinction

between positional binding mechanisms and representational binding — central to the control analysis — draws directly on their circuit taxonomy.

AI and digital accessibility. A growing body of work applies AI to accessibility tasks — generating alt text, detecting contrast failures, automating WCAG audits. We take the inverse approach: rather than using AI as an accessibility tool, we examine what AI systems actually know about accessibility and when that knowledge emerges.

Methodology

Models

This study uses the Pythia model suite (Biderman et al., 2023): 160M, 410M, 1B, 2.8B, and 6.9B parameter models. Pythia models share consistent architecture and training data across all sizes, isolating scale as the variable of interest. All models were loaded using TransformerLens (Nanda et al., 2022).

One architectural exception: Pythia 1B uses a different configuration than the other models in the suite — 16 layers, 8 heads, $d_{\text{model}}=2048$, $d_{\text{mlp}}=8192$. The other models use 12 heads. This difference is relevant when interpreting per-head binding counts in Experiment 4; comparisons across model sizes account for this asymmetry.

To assess replicability across model families, all four experiments were additionally run on the GPT-2 model suite (Radford et al., 2019): small (117M), medium (406M), large (838M), and XL (1.5B) parameter models. Unlike Pythia, GPT-2 models vary in both layer count and head count across sizes — 12/16/20/25 heads and 12/24/36/48 layers respectively — making direct architectural comparisons across sizes less controlled. GPT-2 was trained on WebText, a dataset of outbound Reddit links, which differs substantially from Pythia’s training corpus (The Pile). Results from both model families are reported and cross-architecture comparisons are discussed further in Section five.

Experiments

Experiment 1: Declarative Knowledge Ten accessibility concept prompts (see Appendix A) were run against each model using greedy decoding (temperature=0, max_new_tokens=10):

```
model = HookedTransformer.from_pretrained("pythia-2.8b")
output = model.generate(prompt, max_new_tokens=10, temperature=0)
```

Prompts covered core accessibility terms (screen reader, alt text, skip link), foundational acronyms (WCAG, ARIA), and general accessibility concepts (focus indicator, keyboard navigation, color contrast, semantic HTML, captions). Responses were evaluated against known correct completions — definitions for concept prompts, purpose statements for functional prompts, and correct expansions for acronym prompts.

Experiment 2: Evaluative Knowledge Five code prompts (see Appendix A) were run against each model (temperature=0, max_new_tokens=20), asking models to identify accessibility violations in HTML snippets. Prompts covered missing alt attributes, non-semantic interactive elements, empty links, unlabeled inputs, and ambiguous link text. Responses were evaluated against correct accessibility explanations.

Experiment 3: Recognition vs. Generation Perplexity measures how expected a sequence is to the model, defined as:

$$PPL(X) = \exp \left(-\frac{1}{N} \sum_{i=1}^N \log P(x_i \mid x_{<i}) \right)$$

Lower perplexity indicates the model finds the text more natural. Comparing perplexity for a correct and incorrect definition across model sizes reveals whether recognition precedes generation. (see Appendix A)

```
def get_perplexity(model, text):
    tokens = model.to_tokens(text)
    logits = model(tokens)
    log_probs = torch.nn.functional.log_softmax(logits, dim=-1)
    token_log_probs = log_probs[0, :-1, :].gather(
        1, tokens[0, 1:].unsqueeze(1)).squeeze()
    return torch.exp(-token_log_probs.mean()).item()
```

Experiment 4: Attention Pattern Analysis Attention weights were extracted across all layers and heads for the prompt “A screen reader is” using TransformerLens’s activation cache:

```
logits, cache = model.run_with_cache(prompt)
attention = cache["pattern", layer] # shape: [heads, seq, seq]
```

Token indices were verified for each model prior to analysis:

```
tokens = model.to_str_tokens(prompt)
print(list(enumerate(tokens)))
```

Binding strength between tokens t_i (“reader”) and t_j (“screen”) was measured as the scalar attention weight:

$$b(t_i, t_j) = A_{ij}^{(l,h)}$$

where $A^{(l,h)}$ is the attention weight matrix at layer l , head h , extracted from the activation cache. Heads were classified into binding tiers based on $b(t_i, t_j)^{(l,h)}$:

$$H_{\text{strong}} = \{(l, h) : b(t_i, t_j)^{(l,h)} \geq 0.5\}$$

$$H_{\text{moderate}} = \{(l, h) : 0.2 \leq b(t_i, t_j)^{(l,h)} < 0.5\}$$

Heads with $b(t_i, t_j)^{(l,h)} < 0.2$ were treated as background noise and excluded from analysis. Results report $|H_{\text{strong}}|$ as the primary binding count.

This experiment was run for Pythia 2.8B and 6.9B. Additional binding pairs (alt text, skip link) were analyzed at 2.8B specifically, as this was the first model to demonstrate consistent declarative knowledge across Experiment 1. For GPT-2, binding analysis was run across all four model sizes for screen reader, with additional compound pairs (alt text, skip link) analyzed at XL. Full attention binding data for all model sizes is available in the project repository.

Reproducibility

All experiments were run with temperature=0 for deterministic outputs on Google Colab with an A100 GPU. Greedy decoding was chosen for mechanistic work to ensure deterministic outputs; stochastic decoding may surface partial knowledge that greedy decoding suppresses and is noted as a direction for future exploration. Full notebooks are available at <https://github.com/trishasalas/how-models-think>

Results

Experiment 1: Declarative Knowledge

A correct response demonstrates that the model has encoded the accessibility concept at the scale tested; incorrect or partial responses indicate the concept has not yet emerged.

Models were tested on ten accessibility concept prompts across all five Pythia sizes. Results show a sharp threshold pattern for core accessibility terms, with acronyms behaving differently.

Correct = matches known definition; Partial = incomplete or imprecise; Incorrect = wrong, off-topic, or loops

Prompt	160M	410M	1B	2.8B	6.9B
A screen reader is	×	×	≈	✓	≈
WCAG stands for	×	×	×	×	✓
A skip link is	×	≈	×	✓	×
The purpose of alt text is	×	×	×	✓	✓
ARIA stands for	×	×	×	×	×
A focus indicator is	×	×	×	×	×
Keyboard navigation allows	×	×	×	×	×
Color contrast is important because	×	×	×	×	×
Semantic HTML helps	×	×	×	×	×
Captions are used for	×	≈	×	≈	×

Several patterns emerge from this data.

Threshold at 2.8B. Screen reader, skip link, and alt text all show correct or near-correct responses first at 2.8B. Below this threshold, responses are either wrong or incomplete.

Screen reader shows non-monotonic behavior. The 2.8B model produces “reads aloud the text,” correctly capturing the auditory output purpose. The 6.9B model produces “reads text on a computer screen,” dropping “aloud.” The larger model is more verbose but less precise on the critical detail. This mirrors the skip link result, where 2.8B is correct and 6.9B regresses. Capability near emergence thresholds can be unstable.

WCAG emerges at 6.9B; ARIA never emerges. Both are foundational accessibility acronyms. WCAG requires 6.9B parameters to expand correctly. ARIA fails at every scale tested — at 160M it produces gibberish, at 410M it loops, and from 1B onward it produces confident but wrong expansions (“Artificial Replacement of a Human Being,” “A Rational Approach to In-

formation and Automation,” “Association of Research Libraries in Africa”). Confidence increases with scale; accuracy does not. This suggests ARIA is rare enough in training data that even 6.9B parameters cannot reliably encode it, while increased scale produces more fluent confabulation rather than correct recall.

General accessibility concepts fail across all scales. Focus indicator, keyboard navigation, color contrast, and semantic HTML produce generic responses at every model size. These terms appear in web-scale training data but without accessibility-specific context. The models complete the prompts plausibly without encoding accessibility meaning.

Replication: GPT-2

The same ten prompts were run across GPT-2 small (117M), medium (406M), large (838M), and XL (1.5B).

Prompt	Small	Medium	Large	XL
A screen reader is	×	×	×	≈
WCAG stands for	×	×	×	✓
A skip link is	×	×	×	×
The purpose of alt text is	×	×	≈	≈
ARIA stands for	×	×	×	×

The core findings replicate directionally. WCAG emerges at XL (1.5B) — a lower parameter count than Pythia’s 6.9B, consistent with WCAG appearing more densely in WebText’s Reddit-sourced content around web standards discussions. Screen reader never fully emerges in GPT-2; even at XL the model produces a partially correct response missing the critical “aloud” detail. ARIA fails at every scale tested in both model families. The declarative–evaluative gap and the pattern of sparse accessibility terms failing at all scales are consistent across architectures.

Experiment 2: Evaluative Knowledge

A correct response requires the model to identify the specific accessibility violation, not just complete the code structure plausibly.

Models were tested on five code prompts requiring identification of accessibility violations. This tests whether models can apply accessibility knowledge, not just produce definitions.

Correct = identifies the accessibility violation accurately; Partial = identifies some issue but not the core violation; Incorrect = wrong, off-topic, or loops

Prompt	160M	410M	1B	2.8B	6.9B
<code></code> missing what	×	×	×	×	×
<code><div></code> with onclick not accessible because	×	×	×	×	×
Problem with <code><a</code> <code>href='#'></code>	×	≈	×	×	×
<code><input type='text'></code> needs a	×	×	×	×	×
'Click here' button is bad because	×	×	×	✓	✓

There is a clear gap between declarative and evaluative knowledge. The 2.8B model correctly defines alt text but cannot identify that `<img src='photo.jpg'>` is missing one — it repeats the prompt and stalls at every scale tested. The question explicitly asks what is missing; no model answers “alt text.”

The only evaluative success is the “Click here” prompt, which succeeds at 2.8B and 6.9B. Ambiguous link text may be more common in training data as a named anti-pattern than missing alt attributes.

The `<div>` onclick responses show a trajectory worth noting. Responses become more specific with scale — from loops at 160M to “not a form control” at 6.9B — without arriving at the correct explanation (keyboard inaccessibility, missing role and tabindex). The 6.9B answer is the closest, pointing toward interactivity expectations, but it does not identify the actual problem. This pattern reflects a broader limitation: models complete code structurally before reasoning semantically. Syntactic plausibility and semantic correctness are separable capabilities, and scale closes the gap only partially.

Replication: GPT-2

The same five code prompts were run across all GPT-2 sizes. The declarative–evaluative gap replicates fully. No GPT-2 model identified missing alt text at any scale. “Click here” emerged as a partial success at large (838M) — earlier by parameter count than Pythia’s 2.8B — but regressed at XL, consistent with the instability observed near emergence thresholds in both model families. The evaluative gap does not close at any scale tested in either architecture.

Experiment 3: Recognition vs. Generation

A preference flip — where the model assigns lower perplexity to the correct definition than the incorrect one — indicates recognition has emerged at that scale.

Perplexity measures how expected a sequence is to the model. Lower perplexity means the model finds the text more natural. Testing whether models assign lower perplexity to a correct definition than an incorrect one reveals whether recognition precedes generation. This experiment was extended to three accessibility compounds to assess whether the recognition-before-generation pattern generalizes.

Model	Screen Reader	Alt Text	Skip Link
160M	Wrong 2.6x	Wrong 1.2x	Wrong 1.9x
410M	Wrong 1.2x	Correct 2.8x	Wrong 1.7x
1B	Correct 2.2x	Correct 1.9x	Wrong 1.3x
2.8B	Correct 4.0x	Correct 1.9x	Wrong 1.1x
6.9B	Correct 3.0x	Correct 3.1x	Wrong 1.8x

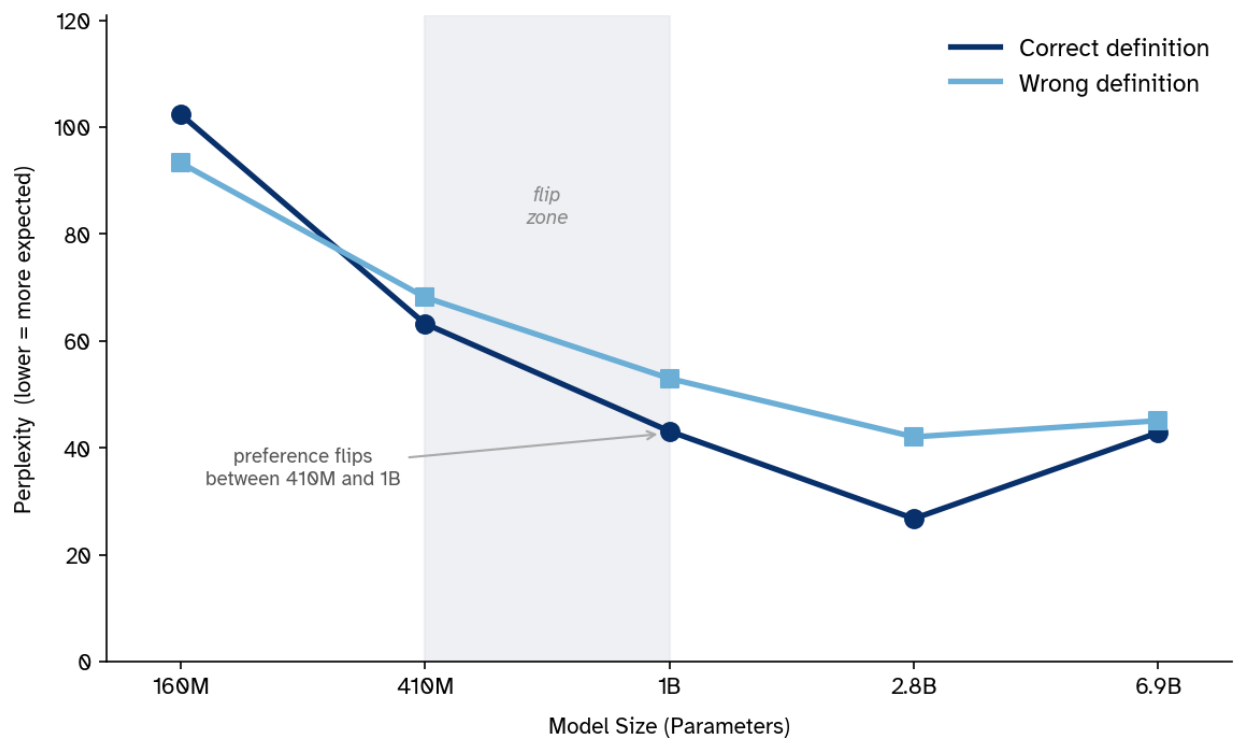


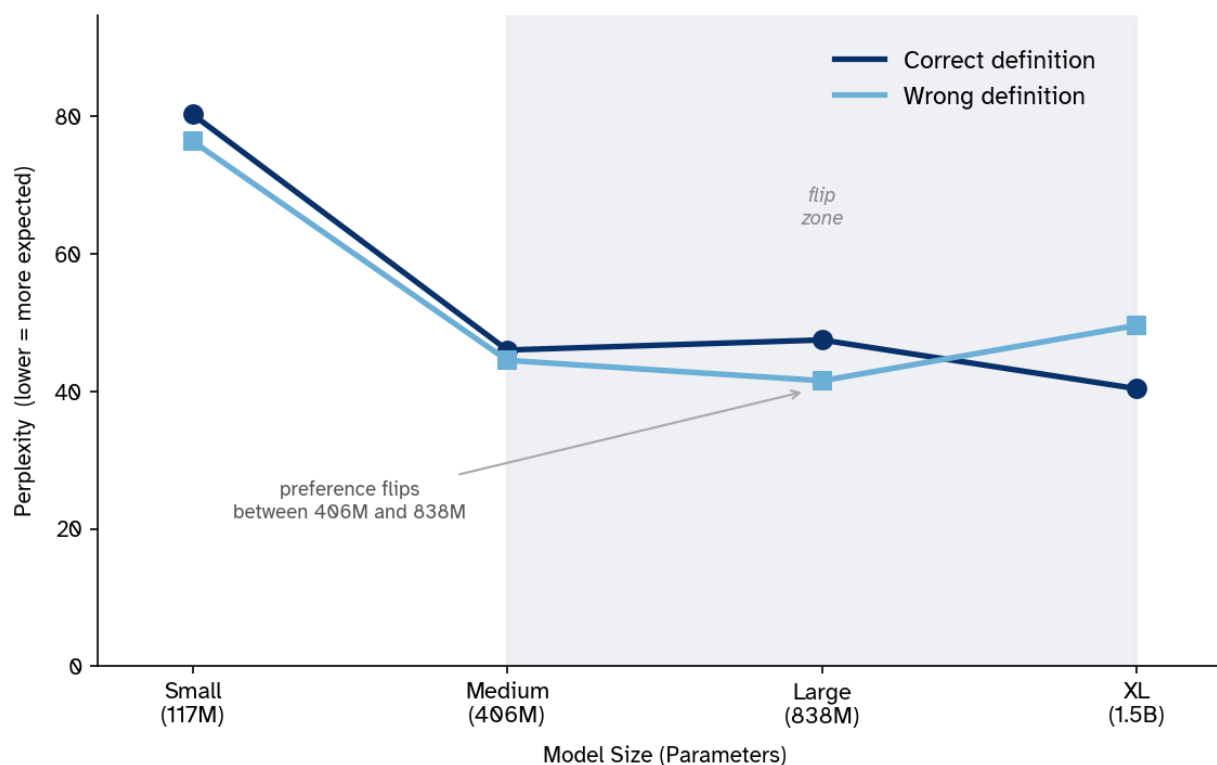
Figure 2: Correct definition perplexity falls below wrong definition perplexity between 410M and 1B parameters in Pythia, indicating the model finds the correct definition more expected before it can generate it.

The preference for screen reader flips between 410M and 1B — recognition precedes generation by one model size. Alt text flips earlier, between 160M and 410M, suggesting it is more densely represented in training data. Skip link never flips; the model finds the incorrect definition more natural at every scale, consistent with the behavioral instability observed in Experiment 1.

The recognition-before-generation pattern holds for two of three compounds tested. Skip link remains the exception across both perplexity and declarative experiments, indicating that some concepts may achieve surface-level generation without the underlying representational grounding that perplexity preference reflects.

Replication: GPT-2

Model	Screen Reader	Alt Text	Skip Link
Small (117M)	Wrong 1.1x	Correct 1.7x	Wrong 1.3x
Medium (406M)	Wrong 1.1x	Correct 1.8x	Wrong 1.4x
Large (838M)	Correct 2.0x	Correct 1.6x	Wrong 2.5x
XL (1.5B)	Correct 2.6x	Correct 2.1x	Wrong 1.8x



Correct definition perplexity falls below wrong definition perplexity between 406M and 838M parameters in GPT-2, replicating the perplexity preference flip observed in Pythia at a comparable scale threshold.

The screen reader preference flips between medium (406M) and large (838M) in GPT-2 — a remarkably similar parameter range to the Pythia flip between 410M and 1B, despite different architectures and training corpora. Alt text is correct-preferring from small in GPT-2, earlier than in Pythia, consistent with its denser representation in web-focused training data. Skip link never flips in either model family. The recognition-before-generation pattern for screen reader replicates across architectures.

Experiment 4: Mechanistic Analysis of Compound Term Binding

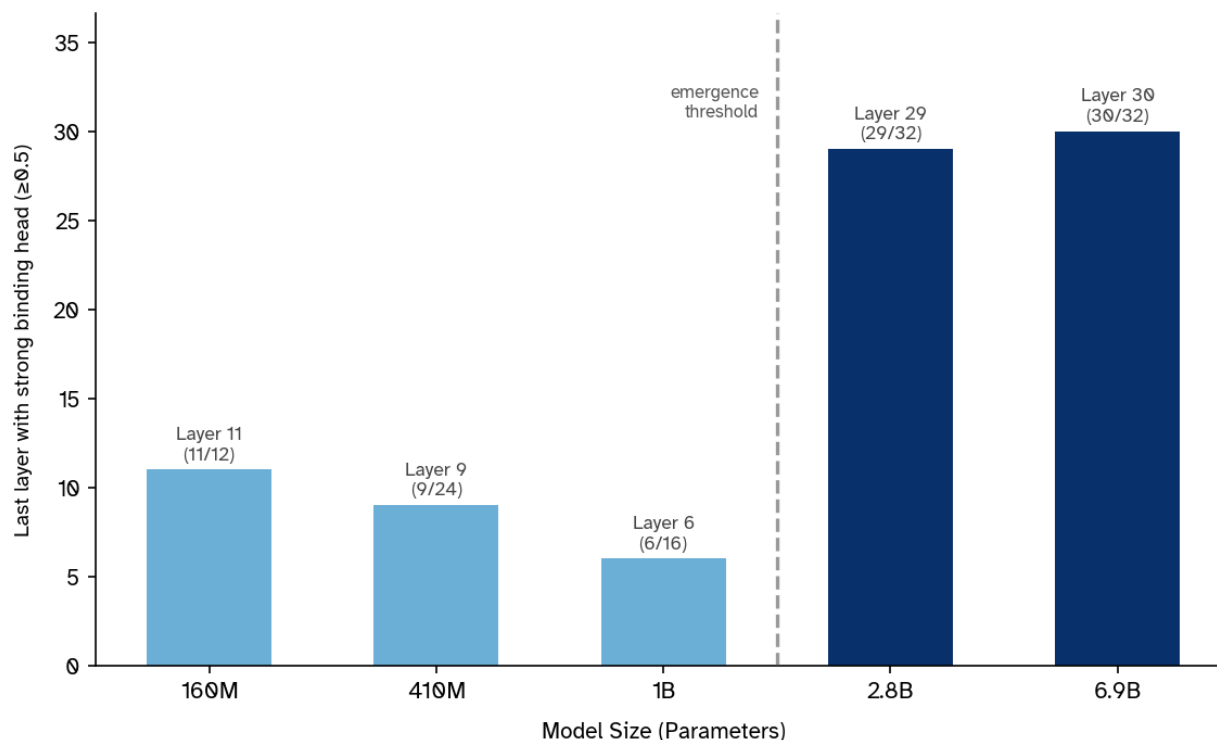
Sustained strong binding into late network layers, rather than early-layer binding alone, is the signal of interest.

Attention pattern analysis across all five Pythia models examines whether models treat “screen reader” as a compound concept or as two independent tokens, and how this binding pattern relates to behavioral emergence.

For each model, attention weights from “reader” to “screen” were extracted across all layers and heads. Heads with weight above 0.5 are considered strong binding; the full data is available in the project repository.

Model	Layers	Strong (0.5+)	Early (L0-3)	Last Layer
160M	12	10	7	11
410M	24	22	16	9
1B	16	11	10	6
2.8B	32	25	20	29
6.9B	32	37	28	30

Note: 1B’s architectural difference (8 heads vs 12) affects raw head counts; see Methodology.



The last layer containing a strong binding head (attention score ≥ 0.5) drops through 160M, 410M, and 1B before jumping sharply at 2.8B, coinciding with the emergence threshold. Models above the threshold show binding heads persisting into the final layers.

Several patterns emerge from this data.

Early layers dominate across all models. Between 70-91% of strong binding occurs in layers 0-3 regardless of model size. Compound term binding is established early in the forward pass at every scale tested.

Strong head count scales with model size. 160M has 10 strong binding heads; 6.9B has 37. 1B is the expected outlier given its different architecture.

Last strong layer tracks behavioral emergence. Below the 2.8B emergence threshold, strong binding drops off early — layer 11 for 160M, layer 9 for 410M, layer 6 for 1B. At 2.8B and above, strong binding persists deep into the network — layers 29 and 30 for 2.8B and 6.9B respectively. The models that cannot correctly define screen reader do not sustain compound binding through the network. The models that can, do.

2.8B is the inflection point. Total heads above threshold jumps from 28 (1B) to 101 (2.8B) — a 3.6x increase that coincides exactly with the behavioral emergence threshold identified in Experiment 1. Early binding is not representational; late binding is. The presence of sustained late-layer binding appears to be a mechanistic bottleneck for concept emergence.

All six models show strong binding in layers 0-3, including 160M which cannot produce a correct definition. Whether early-layer binding at small scales reflects genuine compound representation or proximity effects cannot be determined from attention weights alone and is noted as a limitation.

Control Experiment: Ruling Out Proximity Effects

To test whether the binding signal reflects compound concept encoding rather than simple token adjacency, attention weights were measured between non-compound token pairs at 2.8B. Two conditions were tested: adjacent function words (“and then”) and an adjacent modifier-noun pair without disambiguation (“cold water”). Both conditions produced strong early-layer binding, consistent with the known behavior of previous-token heads (Ols-son et al., 2022) — a class of induction circuit components that attend systematically to the immediately preceding position regardless of content.

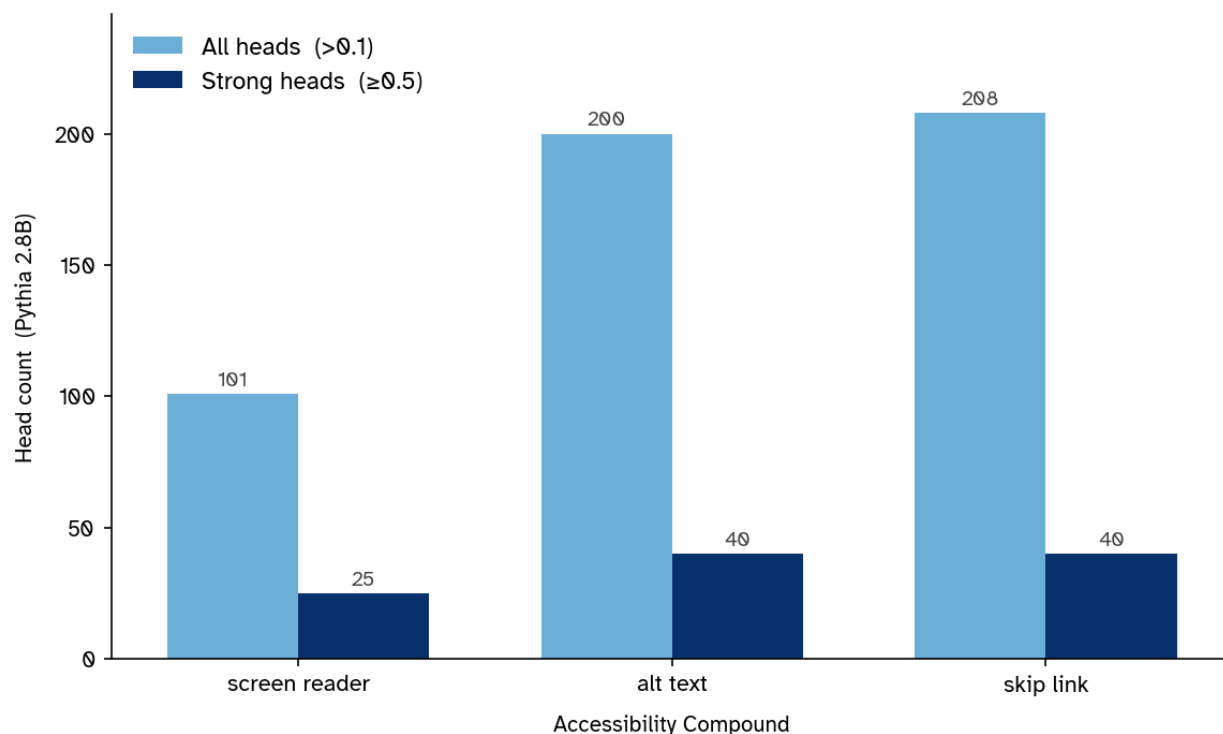
This reframes the control comparison. Early-layer binding is not specific to accessibility compounds; it reflects general positional mechanisms present across token types. The meaningful signal is the distribution and persistence of binding across the full head population. Accessibility compounds at 2.8B recruit 101-208 heads above threshold with strong binding persisting to layers 29-30. Function word pairs produce early-layer binding without the same deep-network persistence. The sustained late-layer binding pattern, rather than raw head count, appears to differentiate accessibility compound binding from general adjacent-token binding. A complete characterization requires systematic comparison across a broader set of controls and is noted as a direction for future work.

Binding Generalizes Across Accessibility Compounds

Having ruled out proximity effects, attention binding was measured for two additional accessibility compound terms at 2.8B: alt text and skip link.

Compound	Total heads >0.1	Strong ($0.5+$)	Top score
screen reader	101	25	0.9909
alt text	211	49	0.9856
skip link	158	32	0.9816

All three compounds show the same pattern of early-layer concentration with deep-network persistence at 2.8B, with top binding scores above 0.98 in each case.



Attention head counts for three accessibility compounds in Pythia 2.8B, showing all heads (attention score >0.1) and strong heads (≥ 0.5). Screen reader activates fewer heads overall, while alt text and skip link show comparable strong head counts despite differences in total activation.

This rules out a compound-specific explanation. The binding pattern is not an artifact of how “screen reader” tokenizes or how frequently it appears in training data. It is a general property of accessibility compound terms at the 2.8B emergence threshold. The models that can define these concepts correctly show robust, distributed binding across many heads and many layers. The models that cannot show weak binding that drops off early.

The binding pattern strengthens the case for a general mechanistic threshold at this scale.

Replication: GPT-2

Attention binding was measured across all four GPT-2 model sizes for screen reader, with additional compounds (alt text, skip link) analyzed at XL.

Model	Total >0.1	Strong (0.5+)	Last Strong Layer	Max Layers
Small (117M)	41	6	4	12
Medium (406M)	66	12	10	24
Large (838M)	146	34	15	36
XL (1.5B)	199	37	17	48

The strong head count jump between medium and large (12 to 34) coincides with the perplexity flip observed in Experiment 3, replicating the Pythia pattern in a different architecture. The last strong layer advances from 10 to 15 at the same transition — consistent with deep-network persistence as a correlate of recognition-level emergence.

At XL, binding was measured for all three compounds:

Compound	Total >0.1	Strong (0.5+)	Last Strong Layer
Screen reader	199	37	17
Alt text	244	34	19
Skip link	253	49	43

One divergence from Pythia is notable. GPT-2 XL’s last strong layer as a proportion of total network depth is substantially shallower than Pythia 2.8B — approximately 35% vs 91%. This may reflect differences in training data composition and architectural differences. Pythia’s training corpus (The Pile) includes technical documentation, Stack Exchange, and academic text where accessibility terminology appears in consistent, well-formed contexts. GPT-2’s WebText corpus is dominated by general web content where the same terms appear more diffusely. The shallower binding depth in GPT-2 may account for the weaker and less stable behavioral emergence observed in Experiment 1.

Discussion

We examined accessibility concept emergence across the Pythia model suite using mechanistic analysis, perplexity-based recognition, and behavioral evaluation. The results converge on a consistent picture with several implications for both accessibility tooling and mechanistic interpretability research.

The 2.8B Threshold Is Meaningful, Not Arbitrary

The behavioral findings show a sharp threshold at 2.8B parameters for core accessibility concepts. This is not a smooth gradient — 1B partially retrieves ‘reads text from a screen’ while 2.8B produces ‘reads aloud the text,’ capturing

the auditory delivery mechanism of a screen reader. The threshold pattern, combined with the 3.6x jump in sustained attention binding heads at 2.8B, suggests a qualitative shift rather than incremental improvement. Below 2.8B, the model has insufficient encoding to support correct generation. At 2.8B, encoding is sufficient to sustain compound binding throughout the network and produce correct behavioral output.

Encoding and Retrieval Are Dissociable

The perplexity results demonstrate that models prefer correct definitions before they can produce them. At 1B, the model assigns lower perplexity to the correct screen reader definition despite generating “reads text from a screen.” This dissociation between recognition and generation suggests that accessibility knowledge exists earlier than the model can articulate it.

This has practical implications: perplexity-based probes may detect accessibility knowledge at smaller scales than behavioral tests suggest.

The Declarative-Evaluative Gap Is Not Closed at Any Scale Tested

Even at 6.9B, models that correctly define accessibility concepts cannot reliably identify violations in code. The single evaluative success, identifying ambiguous link text, may reflect training data frequency rather than genuine accessibility reasoning. The image missing alt attribute failure is particularly telling: a model asked explicitly what is missing produces the prompt back rather than “alt text,” despite correctly defining alt text in declarative tests. Knowing and applying are different capabilities that do not emerge together.

For accessibility tooling practitioners, this gap is the central finding. A model that passes a definition benchmark is not ready for code review tasks.

Acronym Failure Is Systematic

WCAG and ARIA show qualitatively different failure patterns than conceptual terms. Conceptual terms show emergence — partial at 1B, correct at 2.8B. Acronyms either emerge very late (WCAG at 6.9B) or not at all (ARIA at any scale tested). The ARIA failure is particularly informative: hallucinations become more fluent and confident with scale without becoming accurate. At 160M, ARIA produces gibberish. At 6.9B, it produces a coherent but wrong expansion (“Association of Research Libraries in Africa”) with apparent conviction. This is consistent with inverse scaling — where fluency amplifies confident confabulation rather than correct retrieval with increased model size (McKenzie et al., 2023).

This suggests WCAG and ARIA are rare enough in web-scale training data that standard scaling cannot reliably encode them. This pattern is consistent with findings on knowledge frequency effects in parametric memory: model accuracy on factual recall correlates with entity popularity in training data, and rare entities produce confident confabulation rather than correct retrieval (Mallen et al., 2023). Specialized training data or fine-tuning is likely required for reliable acronym expansion in the accessibility domain.

Attention Binding as a Mechanistic Correlate of Emergence

The sustained attention binding pattern at 2.8B and 6.9B — strong binding persisting to layers 29-30 versus dropping off at layers 6-11 in smaller models — provides a mechanistic correlate for behavioral emergence. The models that can define screen reader sustain the compound binding across the full network. The models that cannot drop it early.

Early binding is not representational; late binding is. The presence of sustained late-layer binding appears to be a mechanistic bottleneck for concept emergence: a necessary structural condition without which behavioral competence does not appear. All models, including those that fail behaviorally, show strong binding in early layers. The differentiating factor is whether that binding persists deep into the network.

This finding is correlational, not causal. Whether sustained binding causes correct generation, or both are consequences of a third factor such as MLP encoding depth, cannot be determined from attention weights alone (Geva et al., 2021).

Cross-Architecture Replication

The threshold and binding patterns were replicated on the GPT-2 model suite. The core findings hold across both model families.

The recognition-before-generation pattern for screen reader holds in GPT-2, with the perplexity preference flipping between medium (406M) and large (838M), a parameter range closely matching Pythia’s 410M–1B transition. The declarative–evaluative gap replicates fully: no GPT-2 model identified missing alt text, and ARIA fails at every scale in both families. The attention binding jump between GPT-2 medium and large (12 to 34 strong heads) coincides with the perplexity flip, consistent with deep-network persistence as a correlate of recognition-level emergence.

Notably, GPT-2’s last strong binding layer as a proportion of total network depth is substantially shallower than Pythia 2.8B — approximately 35% vs 91%. Full behavioral emergence is weaker and less stable in GPT-2 across all compounds tested. This may reflect differences in training data composition and architectural differences — GPT-2 uses sequential attention+MLP blocks versus Pythia’s parallel architecture, which may affect how binding reinforcement accumulates across layers. Pythia’s training corpus (The Pile) includes technical documentation, Stack Exchange, and academic text where accessibility terminology appears in consistent, well-formed contexts. GPT-2’s WebText corpus is dominated by general web content where the same terms appear more diffusely. The shallower binding depth in GPT-2 may account for the weaker behavioral emergence: the network encodes the compound relationship but does not reinforce it through enough layers to support reliable generation.

Relationship to Prior Work

Wei et al. (2022) established emergence as a general phenomenon. We treat accessibility concept acquisition as a domain-specific case study: concepts are rare enough to show scale sensitivity, evaluable against clear ground truth, and directly relevant to real-world applications. The results confirm that emergence operates in specialized domains but with domain-specific timing — accessibility acronyms require substantially more scale than conceptual terms, and evaluative capability does not emerge within the range tested.

Prior work on accessibility and LLMs has focused primarily on behavioral evaluation, whether models can identify or remediate violations. We add a mechanistic dimension, not only what models know but how that knowledge is structurally encoded, and at what scale that encoding becomes sufficient for generation. The consistent results in both the Pythia and GPT-2 families strengthen the mechanistic claim that deep-network binding persistence is not limited to a single model family.

The finding that general accessibility concepts (focus indicator, keyboard navigation, color contrast, semantic HTML) fail across all scales in both model families suggests training data frequency is the limiting factor, not model capacity. These terms appear in web-scale data but without the accessibility-specific context required for accurate representation.

Conclusion

We examined accessibility concept acquisition across five Pythia model sizes using mechanistic analysis, perplexity-based recognition, and behavioral evaluation, with replication across the GPT-2 model suite. The findings establish a consistent picture of how and when accessibility knowledge emerges in language models trained on web-scale data.

Core accessibility concepts — screen reader, alt text, skip link — emerge behaviorally at 2.8B parameters in Pythia. Foundational acronyms require more scale (WCAG at 6.9B) or fail entirely (ARIA at all scales tested). General

accessibility vocabulary fails across all model sizes in both model families, suggesting training data frequency as the limiting factor rather than model capacity. Evaluative capability — identifying accessibility violations in code — does not emerge within the range tested, even in models that correctly define the relevant concepts.

The perplexity results show that recognition precedes generation: models prefer correct definitions before they can produce them, with the preference flipping between 410M and 1B in Pythia and between 406M and 838M in GPT-2, a remarkably consistent parameter range across different architectures and training corpora. The attention binding analysis shows that sustained binding of “screen reader” as a compound concept across the full network is a mechanistic correlate of emergence — present at 2.8B and 6.9B in Pythia, absent at smaller scales. The same binding jump replicates in GPT-2 at the transition between medium and large, coinciding with the perplexity flip. Early binding is not representational; late binding is.

For practitioners building accessibility tooling on language models, the central implication is that behavioral success on definition tasks does not predict success on evaluation tasks. The declarative-evaluative gap observed here replicates across both model families and suggests that reliable accessibility code review requires either substantially larger models or domain-specific training data that goes beyond the web-scale distribution.

For emergence researchers, accessibility concepts offer a useful probe domain: rare enough to show scale sensitivity, concrete enough to evaluate against ground truth, and distinct enough from general knowledge to isolate domain-specific acquisition. The ARIA confabulation pattern — increasingly fluent wrong answers with scale — illustrates a failure mode that definition-based benchmarks would not detect.

The mechanistic findings here are correlational. Whether sustained attention binding causes behavioral emergence or reflects a common underlying factor requires causal analysis beyond the scope of this study and is a direction for future work.

Limitations

We report findings from a first investigation into accessibility concept emergence across the Pythia model suite, with replication across GPT-2. Several limitations should be considered when interpreting the results.

Prompt Design

All experiments used completion-style prompts (“A screen reader is,” “WCAG stands for”) rather than instruction or question formats. Prompt framing is known to affect retrieval in language models, and our own findings support that sensitivity. The alt text retrieval experiment demonstrated that the HTML-framed prompt (“In HTML, the alt attribute provides a text description of an”) performed worse than the simpler framing despite testing identical knowledge. Results reported here reflect one prompt design choice and may not generalize to other framings.

The declarative prompts were adapted from attention binding experiments rather than designed independently for behavioral testing. Future work should compare multiple prompt formulations to establish which results are robust across framings.

Evaluative Experiment Prompts

The evaluative experiment used zero-shot code completion prompts. The image missing alt attribute result — where models loop the prompt rather than identify the missing attribute — may reflect document completion behavior rather than a knowledge failure. Models trained on code repositories may interpret `<img src='photo.jpg'>` as the beginning of a code listing and complete accordingly. Instruction-tuned models or explicit question formatting (“What accessibility attribute is missing from this HTML?”) may produce different results. This limitation is noted but not resolved here.

Perplexity Experiment Scope

Recognition versus generation was tested using three sentence pairs across screen reader, alt text, and skip link. The preference flip is a clear finding for screen reader and alt text, but skip link fails to flip in either model family, suggesting the recognition-before-generation pattern does not hold uniformly across all accessibility compounds. A broader set of correct/incorrect pairs across additional concepts would further characterize which compounds follow this pattern and which do not.

Attention Binding Methodology

The proximity control experiment was run at 2.8B only. Control testing revealed that adjacent token pairs — including function word pairs and modifier-noun pairs without disambiguation — produce strong binding at L1H12, which was subsequently confirmed as a previous-token head attending systematically to the immediately preceding position regardless of content. This finding limits the interpretation of L1H12’s consistent appearance across accessibility compounds. Whether other heads show accessibility-specific binding patterns that do not appear in non-compound controls has not been fully characterized and is noted as a direction for future work. Early-layer binding at 160M (including Layer 11 Head 8 at score 1.0) remains ambiguous. Whether this reflects genuine compound representation or positional attention at small scales cannot be determined from attention weights alone.

The compound generalization finding — that alt text and skip link show comparable binding to screen reader at 2.8B — is based on head counts above the 0.1 threshold across all three compounds.

Cross-Architecture Comparison

GPT-2 models vary in both layer count and head count across sizes, making direct per-head comparisons across model sizes less controlled than Pythia’s uniform architecture. The binding depth comparison between GPT-2 and

Pythia is expressed as a proportion of total network depth to account for this, but the architectures are not directly commensurable. Additional replication on architectures with controlled scaling (such as other Pythia-style suites) would strengthen the cross-architecture claims.

Greedy Decoding

All experiments were run with temperature=0 for deterministic outputs. Greedy decoding ensures reproducibility but may suppress partial knowledge that stochastic decoding could surface. Models with partial accessibility encoding may perform differently under sampling-based generation. This is noted as a direction for future exploration.

Hardware and Reproducibility

All experiments were run on Google Colab with an A100 GPU. All notebooks are available in the project repository for independent verification.

Scale Coverage

The scale ladder covers 160M through 6.9B parameters for Pythia and 117M through 1.5B for GPT-2. ARIA fails to emerge at any scale tested in either model family. Whether ARIA would emerge at larger scales is unknown. The evaluative gap — models that define concepts cannot identify violations — was not tested beyond 6.9B. Both findings may reflect limitations of the scale range rather than fundamental properties of these concepts.

Extended Analysis: Elicitation Robustness & Additional Observations

Introduction

The original experiments relied on two prompt types to assess evaluative reasoning. A natural question follows: could the observed gap be an artifact of how we asked rather than a genuine capability boundary? To address this, we expanded to multiple elicitation strategies, varying the prompt structure while holding the underlying concept constant.

Testing was limited to Pythia 1B and 2.8B, the two model scales where the declarative-evaluative gap was originally observed.

Completion Tasks

We used two different elicitation strategies, few-shot structural and bare completion, making sure that there was no accessibility language in either prompt. (*Appendix C, Table 6*) 2.8B completed every accessibility concept correctly except the bare closed captions prompt. 1B struggled across the board but got close on a couple. The control prompts completed correctly on both models, confirming that the differences are specific to accessibility concepts, not general HTML ability.

This establishes that the declarative knowledge is there — the model knows these patterns. The question is what happens when we ask it to reason about them.

Hypothesis-Driven Evaluation

To test whether prompt framing influenced the evaluative failure observed in Experiment 2, three elicitation strategies were introduced: direct questioning, error correction, and Socratic prompting. (*Appendix C, Table 7*)

All three strategies produced the same core outcome. The model could recognize the accessibility concept but failed to identify the violation.

The consistent prompt structures tells us that the declarative-evaluative gap is real, it's robust across multiple elicitation strategies, and it's not an artifact of how we asked. The failure appears to reflect a genuine capability boundary rather than a retrieval problem.

Validation Prompts

A small validation set was introduced to determine whether the observed failures reflected prompt-specific behavior or a broader pattern. (*Appendix C, Table 8*) These prompts produced a range of responses rather than simple pass/fail outcomes. In several cases the model correctly identified the accessibility problem but could not articulate the specific solution.

One response was particularly striking. When asked “What accessibility attribute is missing from this HTML: `<img src='photo.jpg'>?`”, the model responded:

“I have a photo gallery on my website. I want to add an accessibility attribute to the image tag so that the image can be clicked on. I have tried adding the attribute to the img tag, but it doesn't work.”

The model appears to recognize the issue while lacking the structured reasoning needed to resolve it — describing its own gap from inside the gap.

Cross-Model Comparison

To distinguish domain-specific behavior from general capability limits, the same prompt structures were tested on a smaller model (1B).

The 1B model fails broadly across tasks, producing generic or off-topic completions. In contrast, the 2.8B model shows a split profile: strong perfor-

mance on declarative completion tasks but consistent failure on evaluative reasoning.

This contrast localizes the gap. The issue is not general model weakness but a specific failure to transition from declarative knowledge to evaluative reasoning.

Entropy Analysis

We computed mean and last-token entropy on the 3 hypothesis-driven prompts for both 2.8B and 1B

2.8B recognizes “screen reader” territory and locks in confidently — but it’s confident about PARROTING, not about answering. 1B doesn’t even recognize the territory. High entropy corresponds to stalled or echoing completions; intermediate entropy corresponds to fluent but incorrect explanations; low entropy corresponds to confident continuation of the prompt’s rhetorical structure.

The 1B model shows no comparable differentiation. This provides quantitative evidence that different prompt structures push the larger model into distinct predictive states even when the final answer is incorrect.

Model	Prompt	Mean Entropy	Last Token Entropy
1B	Direct Question	4.7761	5.6073
1B	Error Question	5.6830	4.9935
1B	Socratic	4.9666	5.7079
2.8B	Direct Question	4.4127	5.1879
2.8B	Error Question	5.7185	5.5864
2.8B	Socratic	4.5445	4.1660

Extended Binding Analysis

To determine whether the sustained binding pattern continues beyond the original model suite, we extended the attention binding analysis to Pythia 12B. We used the same methodology as the original experiment: threshold ≥ 0.5 for strong binding, ≥ 0.1 for inclusion.

12B Binding Results

- 119 total heads above 0.1 threshold
- 36 strong heads above 0.5 threshold
- Last strong binding layer: 34 (out of 35, zero-indexed) = 97% depth
- Top binding head: Layer 1, Head 15 at 0.9905

The late-layer resurgence finding: Layer 33 has 7 active binding heads (4 above 0.5), including Head 16 at 0.8766. This is the strongest late-layer cluster of any model tested.

Late-layer binding entries scale with model size:

- 1B: 1 late entry
- 2.8B: 4 late entries
- 6.9B: 6 late entries
- 12B: 11 late entries

The specialization finding: % of heads active in binding DECREASES with scale: 31.9% at 160M $\times$ 8.3% at 12B. But the absolute count plateaus around 100-120 for models ≥ 2.8 B. The model figures out how many heads it NEEDS and stops adding more. Larger models use proportionally fewer but more specialized heads.

Cross-model pattern

- Binding is front-loaded in ALL models (65-82% of heads in first 25% of layers) — universal
- Strong heads (>0.9) cluster at 3-10% depth for all models ≥ 1 B — universal

- Late-layer resurgence scales with model size — this is the new finding
- Mean binding scores are stable (~ 0.36 - 0.44) regardless of scale — the STRENGTH doesn't change, the DEPTH does

The key point: 12B extends the published scaling curve. The original paper's bar chart gains a sixth data point. Sustained binding continues to push deeper with scale. The late-layer resurgence is a new observation not in the original paper — a cluster of binding heads re-engaging near the output layers, scaling with model size.

This is consistent with the hypothesis that larger MLP capacity sustains compound representations through deeper network propagation. Direct MLP investigation is future work.

Additional Discussion

The extended analysis was motivated by a straightforward question: could the declarative-evaluative gap observed in the original experiments be an artifact of prompt design? Across 15 prompts, 3 elicitation strategies, 4 accessibility concepts, and a non-accessibility control, the answer is no. The gap persists regardless of how we ask.

But the failure isn't uniform. Entropy analysis reveals that the model enters measurably different internal states depending on the prompt structure, high entropy when it stalls, intermediate entropy when it confabulates and low entropy when it confidently parrots. The model is processing these prompts differently even when the outcome is the same: failure. The Socratic prompt in particular produces a striking confidence signature at 2.8B that warrants circuit-level investigation in future work.

The 12B binding analysis extends the original scaling curve with a sixth data point and introduces a new observation: late-layer resurgence. A cluster of binding heads re-engages near the output layers, and this cluster scales with model size. Larger models also show increased specialization — fewer heads participating in binding overall, but sustaining it deeper through the network.

This is consistent with the hypothesis that wider MLP layers enable sustained propagation of compound representations, though direct investigation of MLP activations is deferred to future work.

Not all concepts benefit equally from scale. Skip link prefers the incorrect definition even at 12B, suggesting that training data representation may impose a floor that scale alone cannot overcome. Similarly, cross-domain confabulation patterns observed for terms like ARIA suggest the phenomenon may not be specific to accessibility, though this paper uses accessibility as its lens. Both directions are left for future investigation.

Appendix A

1. Declarative Prompts

All declarative experiments used completion-style prompts. Models were given the prompt prefix and generated a continuation.

- A screen reader is
- WCAG stands for
- A skip link is
- The purpose of alt text is
- ARIA stands for
- A focus indicator is
- Keyboard navigation allows
- Color contrast is important because
- Semantic HTML helps
- Captions are used for

2. Evaluative Prompts

Evaluative experiments used zero-shot code completion prompts.

- The following code is not accessible because it doesn't have what?
`<img src='photo.jpg'>`
- A `<div>` with `onclick` is not accessible because
- The accessibility problem with `<a href='#'></a>` is
- `<input type='text'>` needs a
- A button that only says 'Click here' is bad because

3. Perplexity Pairs

Each pair consists of a correct and incorrect definition. Perplexity was computed for each sentence independently. Values below 1.0 on the preference ratio indicate the model finds the wrong definition more natural.

Concept

Screen reader

Correct: A screen reader is software that reads text aloud for blind users.

Incorrect: A screen reader is a device for viewing screens.

Alt text

Correct: The purpose of alt text is to provide a textual description of an image for people with visual disabilities.

Incorrect: The purpose of alt text is to make images load faster.

Skip Link

Correct: A skip link is a navigation aid that allows keyboard users to bypass repetitive content.

Incorrect: A skip link is a broken hyperlink that does not load.

Appendix B

Model Architectures

All experiments used base (non-instruction-tuned) model checkpoints.

Pythia Scaling Suite Architecture: parallel attention + MLP blocks.

Table 2

Model	Layers	Heads	d_model	d_mlp
160M	12	12	768	3,072
410M	24	16	1,024	4,096
1B	16	8	2,048	8,192
2.8B	32	32	2,560	10,240
6.9B	32	32	4,096	16,384

GPT-2 Model Family Architecture: sequential attention + MLP blocks.

Table 3

Model	Layers	Heads	d_model	d_mlp
Small (117M)	12	12	768	3,072
Medium (406M)	24	16	1,024	4,096
Large (838M)	36	20	1,280	5,120
XL (1.5B)	48	25	1,600	6,400

Appendix C

1. Completion Prompts

- **Alt text - Few Shot:**

```



<track kind="captions" src="intro.vtt"></video>
<video src="demo.mp4"><track kind="captions" src="demo.vtt"></video>
<video src="lecture.mp4">
```

- **Closed Captions - Bare:**

```
<video src="lecture.mp4">
```

- **Page Title - Few Shot:**

```
<html><head><title>About Us</title></head>
<html><head><title>Contact</title></head>
<html><head>
```

- **Page Title - Bare:**

```
<html><head><meta charset="utf-8">
```

- **Script Source - Bare (control):**

```
<script
```

- **Script Source - Few Shot (control):**

```
<script src="utils.js"></script>
<script src="main.js"></script>
<script src="
```

2. Hypothesis Driven Prompts

- **Direct Question:**

Look at this HTML: `<img src='portrait.jpg'>`.
What specific attribute is missing to make it accessible?

Expected Failure: Model repeats the tag or provides a generic definition of images rather than ‘alt text’.

- **Error Correction:**

Correct the following inaccessible code: `<img src='logo.png'>`.

Expected Failure: Model completes the line with more attributes (like width or height) but ignores the alt attribute.

- **Negative Constraint (Socratic)**

A screen reader is software that reads text aloud.

Why would a screen reader fail to describe this tag:

`<img src='chart.svg'>?`

Expected Failure: Despite the ‘screen reader’ anchor (which is deeply bound at 2.8B), the model cannot link the concept to the lack of an alt attribute.

3. Validation Prompts

- **Direct Completion:**

The following code is not accessible because it doesn’t have what?

`<img src='photo.jpg'>?`

Expected Behavior: Stalls or repeats prompt despite correctly defining ‘alt text’ elsewhere.

- **Instructional Correction:**

What accessibility attribute is missing from this HTML: `<img src='photo.jpg'>?`

Expected Behavior: Failure to answer ‘alt text’.

- **Functional Socratic:**

A screen reader is software that reads text aloud. Why would it fail to describe `<img src='photo.jpg'>?`

Expected Behavior: Model fails to link the ‘screen reader’ definition to the missing ‘alt’ attribute.”

- **Fluent Confabulation (Control):**

ARIA stands for

Expected Behavior: Produces confident but incorrect expansions.

- **Fluent Confabulation (Control):**

HTML stands for

Expected Behavior: Produces confident but incorrect expansions.

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Colophon

This paper is set in Atkinson Hyperlegible Regular, a typeface designed by the Braille Institute to maximize legibility for readers with low vision through distinctive letterforms and numerals that reduce character misrecognition. Choosing it for a paper about accessibility is not incidental.

The PDF was produced using Pandoc with LuaLaTeX. A custom `template.tex` enables `\DocumentMetadata` tagging, producing a document that conforms to both PDF/UA-2 (universal accessibility) and PDF/A-4f (archival) standards. This ensures compatibility with screen readers and other assistive technologies.

This paper was written in collaboration with Claude (Anthropic). The research, methodology, analysis, and conclusions are the author's own.